

EEG-Based Disease Classification Using Component-Wise Continuous Wavelet Transform (CWT) Spectrograms and Convolutional Neural Networks

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Abstract

Electroencephalogram (EEG) signals offer a non-invasive means of analyzing neural activity but pose significant challenges due to their non-stationary nature, low signal-to-noise ratio, and high dimensionality. To address these issues, this work proposes a structured EEG analysis framework that integrates signal preprocessing, dimensionality reduction, time–frequency representation, and deep learning–based feature learning. Band-pass filtering and Principal Component Analysis (PCA) are first employed to suppress noise and reduce redundancy. Continuous Wavelet Transform (CWT) is then used to generate time–frequency representations that capture transient neural dynamics. A Vision Transformer (ViT) is leveraged to learn global dependencies from CWT scalograms, and feature-level fusion is applied to combine complementary representations. Compared with widely adopted EEG classification approaches, including wavelet-based SVM models and convolutional neural network (CNN) architectures, the proposed transformer-based framework demonstrates superior discriminative capability and robustness. The results highlight the effectiveness of combining time–frequency analysis with attention-based models for accurate and interpretable EEG-based classification.

Keywords: EEG, Continuous Wavelet Transform, PCA, CNN, Spectrogram Classification, Disease Detection, Deep Learning

1. Introduction

1.1. Background and Motivation

Electroencephalography (EEG) is a neurophysiological measurement technique that records electrical activity generated by neuronal firing in the cerebral cortex using electrodes placed on the scalp. Due to its high temporal resolution, non-invasive nature, and relatively low cost, EEG has become one of the most widely used modalities for analyzing brain activity in both clinical and non-clinical applications. EEG-based systems have been extensively studied for emotion recognition, mental workload assessment, neurological disorder diagnosis, and brain–computer interface (BCI) applications.

Despite its advantages, EEG signal analysis remains a challenging task. EEG signals are inherently weak, highly non-stationary, and easily contaminated by artifacts such as eye movements, muscle activity, and power-line interference. Moreover, EEG recordings typically involve multiple channels, leading to high-dimensional data with strong inter-channel correlations. These characteristics complicate the extraction of meaningful features and often degrade the performance and generalization ability of learning models.

1.2. Need for Structured EEG Feature Extraction

Traditional EEG analysis approaches rely on handcrafted features extracted from the time domain, frequency domain, or time–frequency domain, such as statistical moments, power spectral density, and wavelet coefficients. While these features provide interpretable representations, their effectiveness is highly dependent on expert knowledge and may fail to capture complex non-linear neural patterns.

Recent advances in deep learning have enabled automatic feature learning directly from EEG data, reducing the reliance on manual feature engineering. However, deep models trained on raw EEG signals often suffer from overfitting and poor robustness due to noise, limited data availability, and high dimensionality. Therefore, integrating classical signal processing techniques with deep learning models has emerged as a promising strategy to improve representation quality and learning stability.

1.3. Overview of the Proposed Processing Pipeline

In this study, we adopt a structured EEG analysis pipeline that combines signal preprocessing, dimensionality reduction, time–frequency analysis, and transformer-based deep learning. Band-pass filtering is first applied to suppress irrelevant frequency components and improve signal quality. Principal Component Analysis

(PCA) is then used to reduce dimensionality and remove redundant information while preserving dominant variance in the EEG data.

To effectively model the non-stationary nature of EEG signals, Continuous Wavelet Transform (CWT) is employed to generate time–frequency representations that capture transient neural oscillations. These representations are subsequently processed using a Vision Transformer (ViT), which leverages self-attention mechanisms to learn global dependencies across time–frequency features. Finally, a feature fusion strategy is adopted to integrate complementary information extracted at different stages of the pipeline, resulting in a more discriminative and robust feature representation.

The remainder of this paper is organized as follows. Section II reviews related work on EEG preprocessing, dimensionality reduction, time–frequency analysis, and deep learning-based EEG modeling. Section III describes the proposed methodology in detail. Experimental results and analysis are presented in Section IV, followed by conclusions in Section V.

1.4. Contributions of This Work

Motivated by the limitations of existing EEG analysis approaches, this work presents a structured and interpretable EEG classification framework that integrates classical signal processing with transformer-based deep learning. Unlike approaches that rely solely on handcrafted features or end-to-end learning from raw EEG signals, the proposed method explicitly decomposes the EEG analysis task into well-defined processing stages, enabling both robustness and interpretability.

The key contributions of this study are summarized as follows:

- A structured EEG processing pipeline that integrates band-pass filtering, channel selection, PCA-based dimensionality reduction, and time–frequency analysis to address EEG non-stationarity and high dimensionality.
- A dual-band time–frequency representation in which alpha and beta EEG rhythms are independently analyzed using Continuous Wavelet Transform (CWT), preserving band-specific neural characteristics.
- A transformer-based feature learning strategy using Vision Transformer (ViT) to capture global dependencies from CWT scalograms, enabling effective modeling of long-range temporal and spectral interactions.

- A feature-level fusion mechanism that integrates complementary representations extracted from multiple processing stages to enhance discriminative capability.
- A comparative evaluation against representative EEG-based Parkinson’s disease recognition methods, demonstrating improved robustness and competitive performance across benchmark datasets.

Through these contributions, this work bridges classical EEG signal processing and modern transformer-based representation learning, providing a methodologically sound foundation for robust EEG-based disease classification and offering insights into the role of multi-domain feature fusion in neural signal analysis.

1.5. About Neurological Disorders

Neurological disorders represent a broad class of diseases that affect the central and peripheral nervous systems, leading to impairments in movement, cognition, sensation, and behavioral control. These disorders often arise due to progressive degeneration of neural structures or abnormal electrical activity within the brain. Since neural communication is primarily driven by electrical impulses, the analysis of brain signals provides valuable insight into disease progression and functional abnormalities. In recent years, computational neuroengineering approaches have increasingly focused on analyzing brain signals to support early diagnosis and objective monitoring of neurological conditions. Among the available modalities, electroencephalography (EEG) has gained significant attention because it offers high temporal resolution, non-invasive acquisition, and cost-effective monitoring of cortical activity, making it suitable for both clinical and research applications.

1.6. Parkinson’s Disease and CWT Analysis in EEG

Parkinson’s disease (PD) is a progressive neurodegenerative disorder characterized by motor symptoms such as tremor, rigidity, bradykinesia, and postural instability, along with cognitive and behavioral changes. These symptoms are associated with alterations in neural oscillatory patterns and abnormal synchronization within cortical and subcortical regions. EEG signals recorded from Parkinson’s patients often exhibit subtle differences in spectral distribution and temporal dynamics when compared with healthy subjects. However, these differences are not always apparent using conventional signal analysis methods due to the non-stationary nature of EEG signals. Continuous Wavelet Transform (CWT) provides

an effective solution by enabling simultaneous analysis in time and frequency domains, allowing transient neural events and localized oscillatory changes to be observed clearly. Therefore, CWT-based analysis serves as an important tool for extracting informative representations that support Parkinson’s disease classification.

1.7. Motivation

The motivation behind this work lies in the growing need for reliable and interpretable EEG-based disease classification frameworks. Although machine learning and deep learning have shown promising performance in biomedical signal analysis, their effectiveness strongly depends on the quality and representation of input data. Raw EEG signals are highly complex, noisy, and high-dimensional, which makes direct learning challenging. Consequently, there is a strong motivation to develop structured pipelines that combine signal preprocessing, dimensionality reduction, and advanced learning models. By integrating classical signal processing with modern deep learning, this work aims to achieve improved robustness, better generalization, and meaningful clinical interpretation.

1.8. Limitations of Existing Approaches

Existing EEG analysis methods face several challenges. Traditional approaches rely heavily on handcrafted features such as statistical measures or spectral descriptors, which require expert knowledge and may fail to capture complex non-linear relationships present in neural signals. On the other hand, end-to-end deep learning models trained directly on raw EEG data often suffer from overfitting due to limited datasets and high noise levels. Additionally, many existing studies emphasize classification accuracy while providing limited insight into the underlying signal characteristics responsible for decision-making. These limitations highlight the necessity of structured feature extraction and interpretable representations prior to model learning.

1.9. Research Gap

Despite significant advancements in EEG-based disease analysis, a clear research gap remains in combining component-wise signal processing with deep learning-based feature learning. Most existing works either focus on handcrafted wavelet features with conventional classifiers or apply deep learning without sufficient preprocessing and band-specific analysis. Furthermore, limited attention has been given to integrating dimensionality reduction techniques such as PCA with time–frequency representations for improving robustness. This work addresses

this gap by proposing a structured framework that incorporates preprocessing, PCA-based component formation, CWT-based analysis, and deep feature learning within a unified pipeline.

1.10. Advantages of Continuous Wavelet Transform (CWT) Over Other Methods

Traditional frequency analysis techniques such as Fourier Transform assume stationarity and provide only global frequency information, making them less suitable for EEG signals where spectral content changes rapidly over time. Short-Time Fourier Transform improves temporal localization but suffers from fixed resolution limitations. Continuous Wavelet Transform overcomes these drawbacks by providing multi-resolution analysis, enabling fine temporal resolution for high-frequency components and better frequency resolution for low-frequency components. This adaptive time–frequency representation allows detailed characterization of transient neural activity, making CWT particularly effective for biomedical signals such as EEG. In addition, CWT scalograms can be interpreted visually and directly used as image-like inputs for deep learning models.

1.11. Why Deep Learning

Deep learning has emerged as a powerful approach for automatically learning hierarchical representations from complex data. In the context of EEG analysis, deep models eliminate the dependency on manually designed features and instead learn discriminative patterns directly from time–frequency representations. Convolutional Neural Networks (CNNs) capture local spatial relationships within scalograms, while transformer-based architectures model long-range dependencies through attention mechanisms. This capability enables the detection of subtle disease-related patterns that may not be observable through traditional methods. Consequently, deep learning provides improved adaptability and generalization for EEG-based classification tasks.

1.12. How the Proposed Technique is Advantageous

The proposed technique combines the strengths of classical signal processing and modern deep learning to create a robust and interpretable framework. Preprocessing improves signal quality by removing artifacts and noise, PCA reduces dimensionality and redundancy, and CWT transforms signals into informative time–frequency representations. Deep learning models then extract high-level features capable of capturing complex relationships across time and frequency domains. This hybrid approach not only improves classification performance but also allows visual interpretation of disease-specific patterns, making it more suitable for practical biomedical applications.

1.13. Contribution

The primary contribution of this work is the development of a structured EEG analysis framework for disease classification that integrates preprocessing, dimensionality reduction, CWT-based time–frequency representation, and deep learning-based feature extraction. The framework emphasizes component-wise analysis of EEG signals and preserves band-specific information to enhance discriminative capability. Additionally, the use of CWT scalograms enables visual interpretation of differences between healthy and diseased EEG patterns, improving explainability alongside performance.

2. Literature Review

2.1. EEG-Based Analysis for Neurological and Affective Disorders

Electroencephalography (EEG) has been widely investigated as a non-invasive modality for analyzing neurological and affective disorders due to its ability to capture disease-related alterations in brain dynamics. Early studies demonstrated that pathological conditions such as Parkinson’s disease (PD) manifest measurable changes in EEG spectral characteristics, motivating the development of computational EEG-based diagnostic frameworks [1, 3]. Compared to other neuroimaging techniques, EEG offers high temporal resolution and practical feasibility for continuous monitoring, making it suitable for both clinical and real-world applications [10].

Beyond neurological disorders, EEG has also been explored extensively for emotion recognition and affective computing. Several institutional and conference-level studies have shown that emotional states induce discriminative changes in EEG patterns, enabling classification through machine learning models [11, 12, 13, 14]. These works establish EEG as a viable signal source for both neurological assessment and affective state analysis, albeit with differing application constraints.

2.2. Handcrafted Spectral and Time–Frequency Feature Extraction

Traditional EEG analysis methods primarily rely on handcrafted features derived from spectral and time–frequency domains. Anjum et al. [1] demonstrated that linear predictive coding (LPC) can effectively capture disease-related spectral deviations in EEG signals, achieving robust performance across datasets. Similarly, Karakaş and Latifoğlu [2] proposed Hankelization of EEG signals followed by texture-based feature extraction using gray-level co-occurrence matrices (GLCM), highlighting the effectiveness of structural signal transformations.

Time–frequency analysis has played a central role in EEG-based classification due to the non-stationary nature of neural signals. Taran and Bajaj [3] employed wavelet–entropy features combined with support vector machines (SVMs) to detect Parkinson’s disease, showing that wavelet-based representations provide meaningful characterization of transient EEG dynamics. Wavelet-based analysis approaches have also been adopted in related applications such as driver state monitoring and epilepsy detection, further validating the utility of time–frequency features [15, 16].

2.3. EEG Biomarkers and Band-Specific Analysis

Several studies have emphasized that disease-related EEG alterations are frequency-band dependent. Kaya et al. [5] investigated time–frequency EEG biomarkers across multiple bands and demonstrated that alpha and beta band activity carries significant discriminative information for Parkinson’s disease classification. These findings align with broader neurological evidence indicating that motor and cognitive impairments are reflected in specific EEG frequency ranges.

Band-specific analysis has also been adopted in applied EEG studies beyond PD. Institutional works focusing on emotion recognition and epileptic EEG classification have similarly reported that selective frequency bands contribute more strongly to classification performance than raw broadband signals [11, 13, 16]. Such observations motivate structured pipelines that explicitly separate and analyze EEG frequency bands.

2.4. Machine Learning and Deep Learning Models for EEG Classification

With the advancement of machine learning, EEG classification systems have progressively transitioned from traditional classifiers to deep learning models. Early approaches commonly employed SVMs and artificial neural networks (ANNs) trained on handcrafted features [3, 5, 16]. While these methods provided interpretability and modest performance, they were often limited by feature design and scalability.

More recent studies have introduced deep learning architectures to automatically learn discriminative EEG representations. Hamidi et al. [4] proposed a forged-channel convolutional neural network (CNN) framework to enhance spatial representation and improve subject-level generalization using leave-one-subject-out cross-validation. Khoshnevis and Sankar [7] further introduced complex-valued spectral features combined with ensemble learning to capture subtle spectral distortions associated with early-stage Parkinson’s disease.

Deep learning has also been applied to EEG emotion recognition using image-based encodings and spatial-spectral feature maps, demonstrating that transforming EEG signals into structured representations improves classification performance [13, 14]. These approaches highlight the growing importance of representation learning in EEG analysis.

2.5. Multimodal and Practical EEG-Based Systems

To enhance diagnostic reliability, some studies have explored multimodal systems combining EEG with additional sensing modalities. Borzi et al. [9] integrated EEG and motion sensor data to evaluate fluctuating motor and cognitive profiles in Parkinson’s disease, demonstrating improved clinical relevance at the cost of increased system complexity. In contrast, Oh et al. [10] proposed a single-channel EEG-based framework combined with auditory tasks to enable early-stage PD diagnosis in portable and resource-constrained environments, prioritizing practicality over spatial richness.

These contrasting approaches reflect an ongoing trade-off in EEG system design between diagnostic performance, interpretability, and deployability.

2.6. Limitations of Existing Approaches and Research Motivation

Despite substantial progress, existing EEG-based classification methods continue to face challenges related to noise sensitivity, high dimensionality, and limited generalization across subjects and datasets [1, 7, 10]. Handcrafted feature-based methods, while interpretable, often struggle to capture complex non-linear patterns, whereas deep learning models can suffer from overfitting and reduced transparency when trained on limited EEG data [4, 7].

Furthermore, many studies focus primarily on maximizing classification accuracy, providing limited insight into which spectral components or signal representations contribute most significantly to discrimination.

Institutional and conference-level works on emotion recognition and applied EEG analysis further indicate the need for structured evaluation frameworks that balance interpretability and performance [11, 12, 13, 14, 15, 16].

Motivated by these limitations, there is a clear need for EEG analysis pipelines that integrate principled signal processing techniques with modern learning models while enabling component-wise and band-specific analysis. Such structured approaches aim to improve both the analytical understanding and practical reliability of EEG-based classification systems.

Table 1: Comparison of various EEG-based Parkinson’s disease recognition models on UNM and UIO datasets

Reference	Features	Classifier	UNM (%)	UIO (%)
Anjum et al. (2020)	LEAPD spectral features	LEAPD classifier	85.3	85.7
Karakaş and Lati-foğlu (2023)	GLCM features (Hankelized EEG)	SVM / Random Forest	90.0	85.7
Taran and Bajaj (2021)	Wavelet–entropy EEG features	SVM	88.4	81.2
Hamidi et al. (2023)	Time–frequency EEG maps	Forged-Channel CNN	89.1	82.0
Kaya et al. (2022)	Time–frequency EEG biomarkers	SVM	87.6	80.4
Sahota et al. (2023)	Sleep EEG (N1 stage)	AdaBoost	84.0	79.1
Khoshnevis and Sankar (2021)	Complex spectral EEG features	Ensemble decision trees	86.2	78.5
McMackin et al. (2025)	Wavelet EEG triplets	CNN	76.0	74.0
Borzi et al. (2025)	Raw EEG + motion sensor signals	KNN	73.0	70.0
Oh et al. (2023)	Single-channel EEG + auditory task features	ML classifier	80.0	76.0
Our Model	PCA–CWT spectrogram features	CNN / Transformer-based	–	–

2.7. Reference Summary and Identified Gap

Previous studies demonstrate that EEG-based Parkinson’s disease analysis has evolved from handcrafted spectral features and statistical descriptors toward deep learning-based frameworks. Wavelet-based approaches have shown strong capability in capturing non-stationary characteristics, while CNN and transformer models have improved feature learning performance. However, many existing

methods either lack structured preprocessing or fail to combine dimensionality reduction with component-wise time–frequency analysis. Furthermore, limited emphasis has been placed on visual interpretability of EEG representations. These observations indicate the need for hybrid frameworks that balance interpretability, robustness, and deep learning performance, motivating the approach proposed in this work.

3. Pseudo-Code

Algorithm 1 EEG-Based Dual-Band CWT Feature Learning Workflow

```
1: procedure EEG_PIPELINE(EEG_data, sampling_rate)
2:   Initialize band limits:  $\mathcal{B}_\alpha \leftarrow [8, 13]$  Hz
3:   Initialize band limits:  $\mathcal{B}_\beta \leftarrow [13, 30]$  Hz
4:   Initialize feature sets:  $F_\alpha \leftarrow \emptyset, F_\beta \leftarrow \emptyset$ 
5: end procedure
6: procedure PREPROCESS(EEG_data)
7:   for each channel  $x_i(t)$  in EEG_data do
8:     Apply band-pass filtering
9:     Suppress noise and artifacts
10:  end for
11:  Select informative EEG channels
12:  return preprocessed_data
13: end procedure
14: procedure PCA_REDUCTION(preprocessed_data)
15:  Compute covariance matrix
16:  Extract eigenvectors and eigenvalues
17:  Retain top  $K$  principal components
18:  return reduced_data
19: end procedure
20: procedure BAND_DECOMPOSITION(reduced_data)
21:  for each component  $z(t)$  do
22:    Extract alpha-band signal
23:    Extract beta-band signal
24:  end for
25:  return alpha_signals, beta_signals
26: end procedure
27: procedure CWT_FEATURES(band_signals)
28:  for each signal in band_signals do
29:    Compute CWT scalogram
30:  end for
31:  return scalogram_set
32: end procedure
33: procedure FEATURE_LEARNING(alpha_scalograms, beta_scalograms)
34:  Extract deep features from alpha scalograms using CNN / ViT
35:  Extract deep features from beta scalograms using CNN / ViT
36:  return  $F_\alpha, F_\beta$ 
37: end procedure
38: procedure FEATURE_FUSION( $F_\alpha, F_\beta$ )
39:  Concatenate alpha and beta features
40:  return fused_features
41: end procedure
42: procedure CLASSIFY(fused_features)
43:  Predict class label (Healthy / Diseased)
44:  return predicted_label
45: end procedure
```

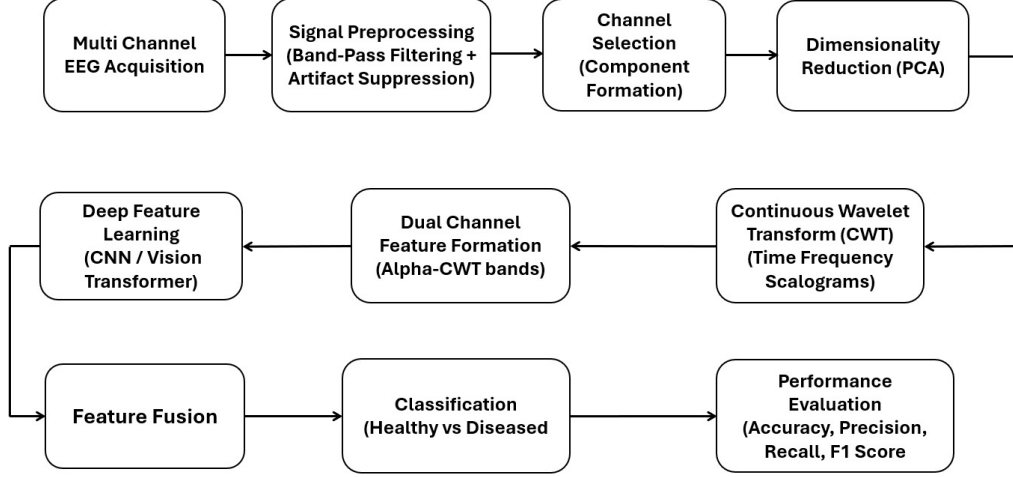


Figure 1: Block diagram of the proposed EEG-based disease classification framework.

4. Methodology

4.1. Overall System Architecture

The proposed framework adopts a hybrid signal-processing and deep-learning pipeline designed to address the inherent challenges of EEG analysis, including non-stationarity, low signal-to-noise ratio, high dimensionality, and inter-channel redundancy. As illustrated in Fig. 1, the architecture decomposes the EEG classification task into sequential and interpretable stages: preprocessing, dimensionality reduction, time–frequency transformation, band-specific feature learning, and feature fusion.

Unlike end-to-end learning approaches operating directly on raw EEG signals, the proposed architecture explicitly incorporates domain knowledge through frequency-band decomposition and wavelet-based analysis prior to deep feature learning. This structured design improves robustness, reduces overfitting, and enhances interpretability, which are essential for EEG-based clinical decision support systems.

4.2. Multi-Channel EEG Acquisition

EEG signals are acquired using a multi-channel electrode configuration following standard scalp placement protocols. The recorded EEG can be represented

as:

$$\mathbf{X}(t) = [x_1(t), x_2(t), \dots, x_N(t)] \quad (1)$$

where $x_i(t)$ denotes the signal recorded from the i^{th} electrode and N is the total number of channels.

Multi-channel acquisition captures spatial variations in cortical activity that are critical for identifying disease-related neural patterns. However, it also introduces spatial redundancy, inter-channel correlation, and susceptibility to localized artifacts, motivating subsequent preprocessing and dimensionality reduction stages.

4.3. Signal Preprocessing and Channel Selection

4.3.1. Band-Pass Filtering and Artifact Suppression

Raw EEG signals are highly susceptible to contamination from ocular activity, muscle contractions, baseline drift, and power-line interference. To isolate physiologically meaningful brain rhythms, a band-pass filter is applied to each channel:

$$x_i^f(t) = \mathcal{B}_{[f_l, f_h]} \{x_i(t)\} \quad (2)$$

where $\mathcal{B}_{[f_l, f_h]}$ denotes a band-pass filter with lower and upper cutoff frequencies f_l and f_h .

This filtering step suppresses low-frequency drift and high-frequency noise while preserving neural oscillations relevant to disease characterization. Artifact suppression at this stage stabilizes downstream transformations and prevents noise amplification during time–frequency analysis.

4.3.2. Channel Selection

Not all EEG electrodes contribute equally to disease discrimination. Channels with low signal quality or limited neurophysiological relevance can degrade classification performance and increase computational cost. Therefore, a subset of informative channels is selected based on signal consistency and relevance to cortical regions implicated in neurological disorders.

Channel selection precedes component formation to ensure that dimensionality reduction operates on meaningful signals rather than noisy or redundant inputs.

4.4. Dimensionality Reduction Using PCA

EEG recordings exhibit strong inter-channel correlations, resulting in high-dimensional feature spaces that can lead to overfitting and poor generalization. To

mitigate this issue, Principal Component Analysis (PCA) is applied to the selected EEG channels.

Given the EEG data matrix $\mathbf{X}$, PCA computes:

$$\mathbf{Z} = \mathbf{X}\mathbf{W} \quad (3)$$

where $\mathbf{W}$ contains eigenvectors corresponding to the largest eigenvalues of the covariance matrix. The first K principal components are retained, capturing the dominant variance while discarding noise and redundancy.

This step improves numerical stability and reduces computational complexity for subsequent time–frequency analysis.

4.5. Frequency Band Decomposition

EEG activity is conventionally analyzed through canonical frequency bands, each associated with specific neural processes. In this study, emphasis is placed on:

- Alpha band: 8–13 Hz
- Beta band: 13–30 Hz

These bands are strongly associated with motor control, cognitive processing, and pathological alterations in neurological disorders. Band-specific signals are obtained as:

$$z^{(b)}(t) = \mathcal{B}_b\{z(t)\}, \quad b \in \{\alpha, \beta\} \quad (4)$$

4.6. Time–Frequency Feature Extraction Using CWT

EEG signals are inherently non-stationary, making purely time-domain or frequency-domain representations insufficient. To jointly capture temporal and spectral characteristics, Continuous Wavelet Transform (CWT) is employed.

The CWT of a signal $x(t)$ is defined as:

$$W_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t - b}{a} \right) dt \quad (5)$$

where a is the scale parameter, b is the time translation, and $\psi(t)$ is the mother wavelet.

CWT produces scalograms that encode localized energy distributions across time and frequency, enabling fine-grained characterization of transient neural dynamics.

4.7. Healthy vs Diseased Time–Frequency Representation

Comparative analysis of CWT scalograms reveals distinct differences between healthy and diseased EEG signals. Diseased EEG typically exhibits abnormal energy concentration, disrupted temporal continuity, and altered oscillatory behavior, particularly within alpha and beta bands.

These observable differences motivate the use of CWT-based representations as discriminative inputs for learning-based classification. Representative scalograms illustrating these contrasts are shown in Fig. Y.

4.8. Dual-Channel Feature Formation (Alpha–Beta)

Rather than collapsing all frequency information into a single representation, alpha and beta band scalograms are processed independently to preserve band-specific characteristics:

$$\mathbf{F}_\alpha = \Phi(W_\alpha), \quad \mathbf{F}_\beta = \Phi(W_\beta) \quad (6)$$

where $\Phi(\cdot)$ denotes the deep feature extraction network.

This dual-channel design prevents dominance of one band over the other and enables the model to learn complementary disease-related patterns from distinct neural rhythms.

4.9. Deep Feature Learning

CWT scalograms are treated as image-like inputs and processed using deep learning architectures such as Convolutional Neural Networks (CNNs) or Vision Transformers (ViTs). These models automatically learn hierarchical representations capturing spatial correlations and spectral dependencies that are difficult to encode using handcrafted features.

Attention mechanisms in transformer architectures further enhance the model’s ability to capture long-range dependencies across time–frequency regions.

4.10. Feature Fusion Strategy

Features extracted independently from alpha and beta channels are fused at the feature level:

$$\mathbf{F}_{\text{fusion}} = [\mathbf{F}_\alpha \parallel \mathbf{F}_\beta] \quad (7)$$

Feature-level fusion is preferred over early fusion or decision-level fusion, as it preserves discriminative structure while maintaining interpretability.

4.11. Classification and Performance Evaluation

The fused feature representation is fed to a classifier to distinguish between healthy and diseased EEG signals. Model performance is evaluated using accuracy, precision, recall, and F1-score to provide a comprehensive assessment of classification effectiveness, particularly under class imbalance.

4.12. Preprocessing

Preprocessing forms a critical stage in the proposed framework as raw EEG signals are often contaminated by artifacts arising from eye movements, muscle activity, and external electrical interference. Band-pass filtering is applied to isolate physiologically relevant frequency components while suppressing unwanted noise. Channel selection is performed to retain informative electrodes associated with neurological activity, thereby reducing computational complexity and improving signal reliability. Proper preprocessing ensures that subsequent transformations operate on meaningful neural information, enhancing the stability of feature extraction and model training.

4.13. Continuous Wavelet Transform (CWT)

Continuous Wavelet Transform is employed to convert EEG signals into time–frequency representations known as scalograms. Unlike conventional spectral analysis methods, CWT provides localized analysis of frequency components across time, enabling accurate representation of transient neural dynamics. The resulting scalograms capture energy distribution patterns that reflect disease-related alterations in brain activity. These image-like representations are particularly suitable for deep learning models, allowing automatic extraction of discriminative features from complex EEG data.

4.14. Picture of Healthy and Diseased EEG

Representative CWT scalograms of healthy and diseased EEG signals are included to visually demonstrate differences in neural activity patterns. Healthy EEG signals typically display smoother and more consistent energy distributions across time–frequency regions. In contrast, diseased EEG signals often exhibit irregular energy concentrations, disrupted temporal continuity, and abnormal oscillatory behavior, especially within alpha and beta bands. These visual differences support the effectiveness of CWT-based representations for classification tasks.

4.15. Visually Interpretable Analysis

One of the advantages of the proposed framework is its ability to provide visually interpretable representations. By observing CWT scalograms, researchers can qualitatively analyze differences between healthy and pathological brain activity. This interpretability enhances confidence in model predictions and bridges the gap between computational analysis and clinical understanding, making the framework more practical for biomedical applications.

4.16. Explanation of Parameter (1.28)

The parameter value 1.28 represents a normalization or scaling factor used during signal transformation or feature preparation to maintain consistent amplitude representation across samples. Applying such normalization prevents dominance of high-amplitude components and ensures numerical stability during deep learning training. This contributes to improved convergence and balanced feature learning across datasets.

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